

Lab 10 Report

Noise Simulation

1. Part 1: LPF AC Noise Analysis

1.1 Output Noise vs. Frequency

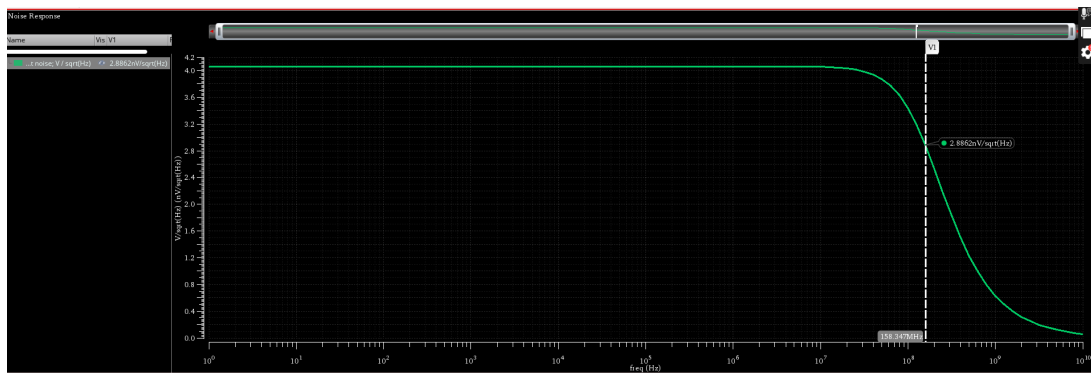


Figure 1: Output voltage noise density $S_{outn}(f)$ vs frequency

AC Noise: Hand Calc vs Simulation

Q: Compare the simulation results (noise density, bandwidth, and rms) with hand analysis.

Parameter	Hand Analysis	Simulation
Noise Density ($nV/\sqrt{Hz}$)	4.07041	4.07133
Bandwidth (MHz)	159.154	158.8
RMS Noise (μV_{rms})	64	64.32

1.2 2. Parametric Sweep of RPAR

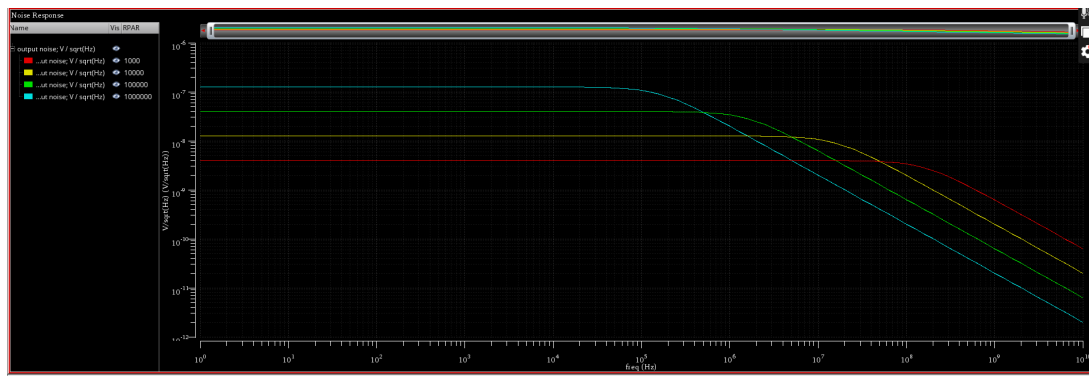


Figure 2: Output noise density for parametric R:1k,10k,100k,1M

Effect of RPAR on Noise

Q: Comment on the overlaid plots. Calculate the rms noise using the calculator for each case and comment on the results.

As RPAR increases, the low-frequency noise density increases ($\propto \sqrt{R}$), but the filter's bandwidth decreases ($\propto 1/R$). Because the total integrated RMS noise for a simple RC filter is given by $\sqrt{kT/C}$, it is independent of the resistance value. Therefore, the RMS noise remains constant across all sweeps.

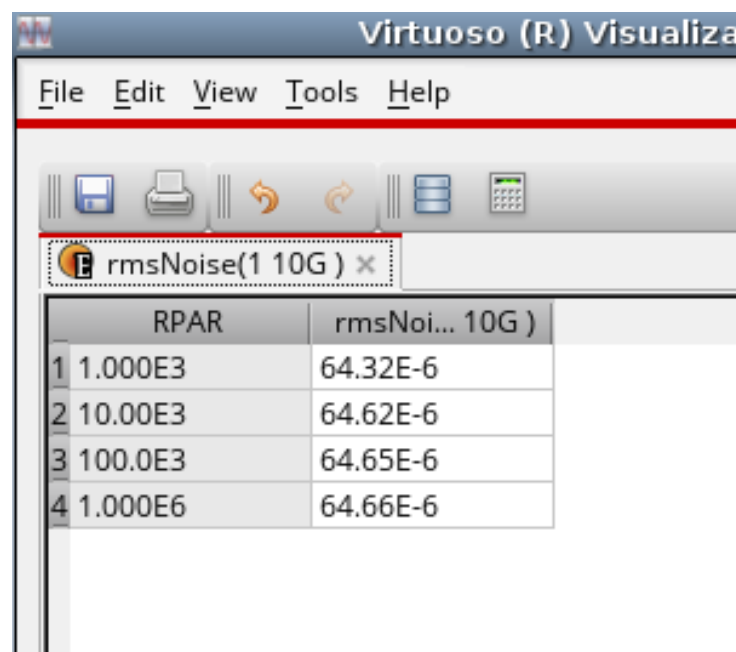


Figure 3: Output rms across all sweeps

2. Part 2: LPF Transient Noise Analysis

2.1 1. Transient Noise Waveform

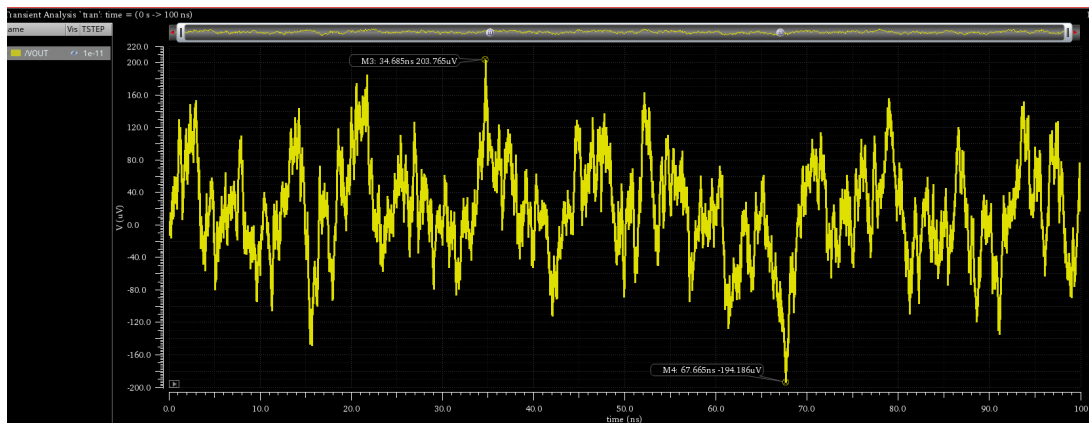


Figure 4: Transient output noise at $TSTEP=TAU/100$

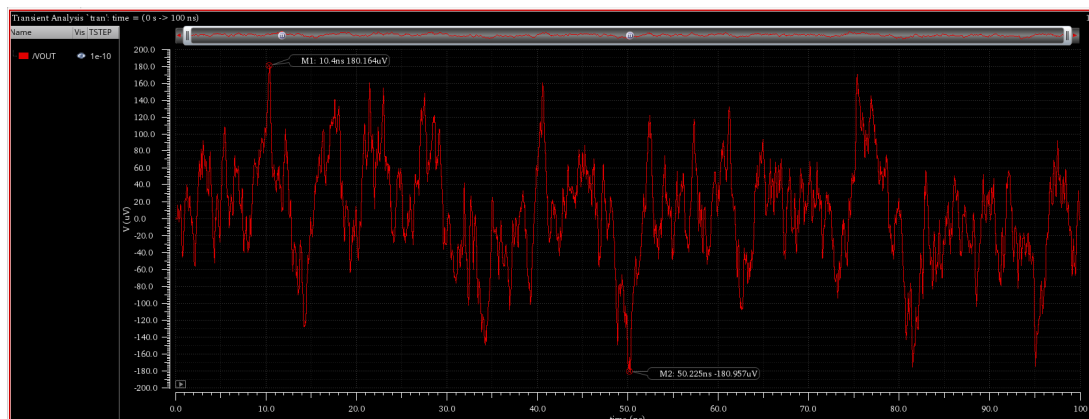


Figure 5: Transient output noise at $TSTEP=TAU/10$

rms(v("/VOUT"?result "tran"))	
TSTEP	rms(v("/... "tran"))
1 100.0E-12	61.59E-6

Figure 6: RMS Noise Simulation ($TSTEP = 100$ ps)

Virtuoso (R) Visualization & Analysis XL table	
File Edit View Tools Help	
cadence	
rms(VT("VOUT"))	
TSTEP	rms(VT(...OUT))
1 10.00E-12	60.88E-6

Figure 7: RMS Noise Simulation ($TSTEP = 10$ ps)

Figure 8: Cadence Transient Noise Simulation Results

Transient RMS Noise & Time Step Dependency

Q: Use the rms function to calculate the rms noise. Compare it to Part 1. Repeat with $TSTEP = TAU/10$. Does the calculated rms noise increase or decrease? Why?

When increasing the time step (e.g., $TSTEP = TAU/100$ to $TAU/10$), the maximum

frequency simulated ($F_{max} = 1/\text{TSTEP}$) decreases. This limits the bandwidth of the integrated noise in the transient simulation, causing the calculated RMS noise to **decrease** because the high-frequency noise components are artificially filtered out by the larger simulator time step.

2.2 2. Multiple Runs Analysis (20 Iterations)

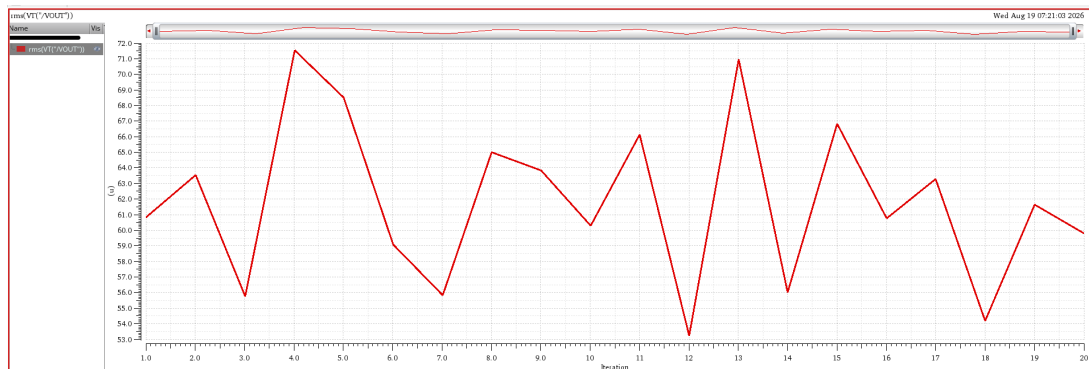


Figure 9: RMS Noise vs iteration at $\text{TSTEP} = \text{TAU}/100$

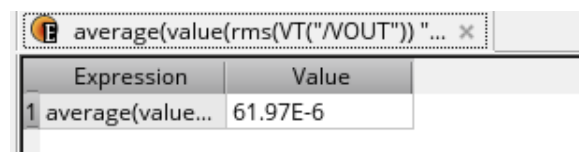


Figure 10: Average rms noise across runs

Average RMS Noise Comparison

Q: Use the calculator to calculate the average rms noise. Compare the calculated value with the rms noise previously obtained in Part 1 and Part 2.

The average RMS noise from the multiple transient runs closely approximates the theoretical integrated RMS noise obtained from the AC noise analysis in Part 1.

3. Part 3: 5T OTA AC Noise Analysis

3.1 1. Thermal Noise vs. Frequency

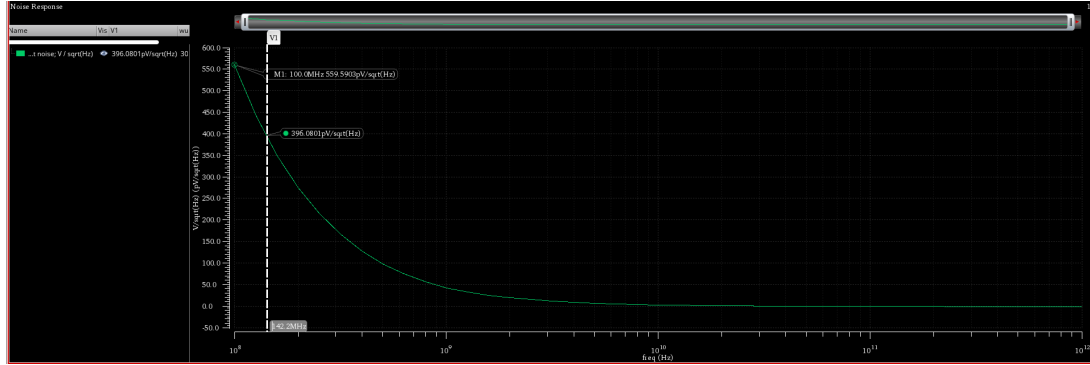


Figure 11: OTA output thermal noise vs frequency

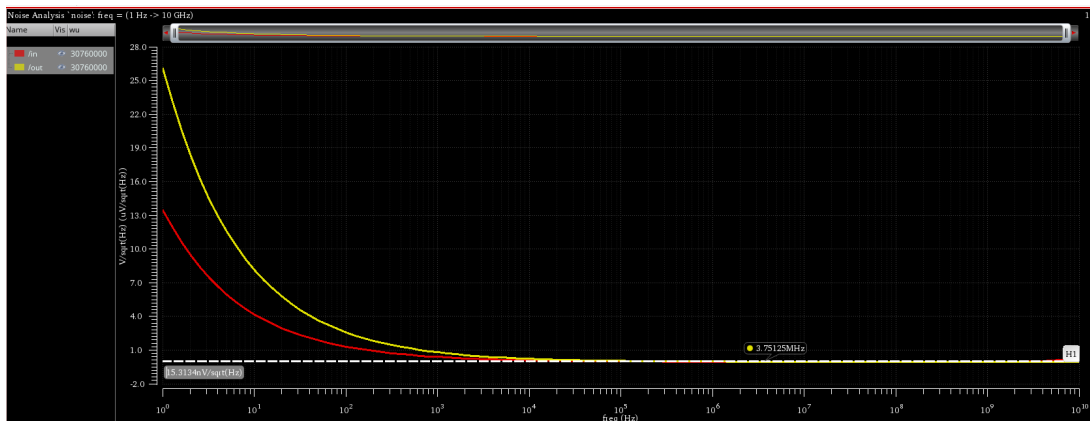


Figure 12: Total OTA output and input referred noise vs frequency, annotated the magnitude of thermal noise voltage density of id

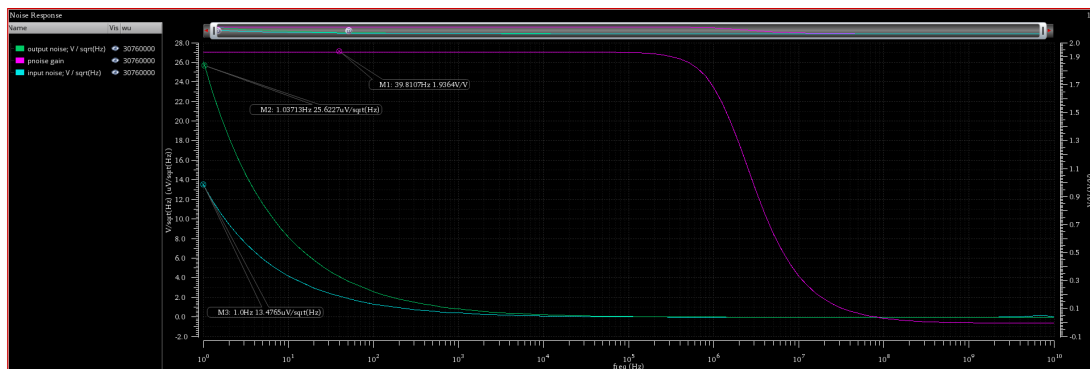


Figure 13: Gain & Total OTA output and input referred noise vs frequency

Device	Param	Noise Contribution	% Of Total
I14.M1.m0	fn	2.85276e-09	38.44
I14.M0.m0	fn	2.75256e-09	37.09
I14.M0.m0	id	4.25334e-10	5.73
I14.M1.m0	id	4.23621e-10	5.71
I14.M5.m0	id	2.91216e-10	3.92
I14.M4.m0	id	2.58408e-10	3.48
I14.M4.m0	fn	2.12982e-10	2.87
I14.M5.m0	fn	2.0315e-10	2.74
Integrated Noise Summary (in V^2) Sorted By Noise Contributors			
Total Summarized Noise = 7.42211e-09			
Total Input Referred Noise = 2.1695e-09			
The above noise summary info is for noise data with wu = 3.076e+07			

Figure 14: Noise Summary

Hand Analysis: Total Integrated Thermal Input-Referred Noise

For a 5-transistor OTA, assuming long-channel devices ($\gamma = 2/3$), the theoretical thermal input-referred noise power spectral density (PSD) is given by:

$$V_{n,in}^2(f) = \frac{8kT\gamma}{g_{m1,2}} \left(1 + \frac{g_{m3,4}}{g_{m1,2}} \right) \quad (1)$$

From the simulated DC operating points, the transconductances are:

- Input pair (M_1, M_2): $g_{m1,2} = 155.9 \mu\text{S}$
- Active load (M_3, M_4): $g_{m3,4} = 102.1 \mu\text{S}$

Assuming room temperature ($T = 300 \text{ K}$) and Boltzmann's constant ($k = 1.38 \times 10^{-23} \text{ J/K}$):

$$V_{n,in}^2(f) = \frac{8 \cdot (1.38 \times 10^{-23}) \cdot 300 \cdot (2/3)}{155.9 \times 10^{-6}} \left(1 + \frac{102.1 \mu\text{S}}{155.9 \mu\text{S}} \right) \quad (2)$$

$$V_{n,in}^2(f) = (1.416 \times 10^{-16}) \cdot (1 + 0.6549) \quad (3)$$

$$V_{n,in}^2(f) \approx 2.34 \times 10^{-16} \text{ V}^2/\text{Hz}$$

To find the total integrated noise, we must multiply the flat thermal PSD by the Effective Noise Bandwidth (ENBW) of the closed-loop system. For a first-order system (typical for an OTA in unity-gain feedback), the ENBW is related to the unity-gain frequency (ω_u):

$$\text{ENBW} = \frac{\pi}{2} f_{3dB} \approx \frac{\pi}{2} f_u = \frac{\pi}{2} \left(\frac{\omega_u}{2\pi} \right) = \frac{\omega_u}{4} \quad (4)$$

Using the ω_u extracted from the Cadence noise summary ($\omega_u = 30.76 \text{ Mrad/s}$):

$$\text{ENBW} = \frac{30.76 \times 10^6 \text{ rad/s}}{4} = 7.69 \text{ MHz} \quad (5)$$

Finally, the total integrated input-referred thermal noise variance ($\overline{V_{in,th}^2}$) and RMS voltage ($V_{in,rms}$) are calculated as:

$$\overline{V_{in,th}^2} = V_{n,in}^2(f) \times \text{ENBW} \quad (6)$$

$$\overline{V_{in,th}^2} = (2.34 \times 10^{-16} \text{ V}^2/\text{Hz}) \times (7.69 \times 10^6 \text{ Hz}) \quad (7)$$

$$\overline{V_{in,th}^2} \approx 1.80 \times 10^{-9} \text{ V}^2 \quad (8)$$

$$V_{in,rms} = \sqrt{\overline{V_{in,th}^2}} = \sqrt{1.80 \times 10^{-9}} \approx 42.4 \mu\text{V} \quad (9)$$

Comparison with Cadence Simulation

The hand analysis calculates only the thermal noise contribution. This theoretical value is compared against the total integrated input-referred noise from the Cadence simulation.

From the Cadence `noise` summary:

- **Simulated Noise Variance:** $2.1695 \times 10^{-9} \text{ V}^2$
- **Simulated RMS Noise:** $\sqrt{2.1695 \times 10^{-9}} \approx 46.58 \mu\text{V}$

Table 1: Integrated Input-Referred Noise: Theory vs. Simulation

Metric	Hand Calc. (Thermal)	Cadence Sim. (Total)	% Diff.
Noise Variance ($\overline{V_{in}^2}$)	$1.80 \times 10^{-9} \text{ V}^2$	$2.17 \times 10^{-9} \text{ V}^2$	$\approx 17.0\%$
RMS Voltage ($V_{in,rms}$)	$42.4 \mu\text{V}$	$46.58 \mu\text{V}$	$\approx 8.9\%$

Discussion of Discrepancies

The simulated RMS noise ($46.58 \mu\text{V}$) exceeds the hand-calculated value ($42.4 \mu\text{V}$) due to two primary factors:

1. **Flicker Noise ($1/f$):** The hand calculation only accounts for wideband thermal noise. The Cadence simulation includes flicker noise, which the log shows is dominant in the NMOS input pair (M_0, M_1), increasing the total integrated variance.
2. **Excess Noise Factor (γ):** The theoretical model uses the long-channel approximation ($\gamma = 2/3$). Short-channel effects in the foundry models typically increase γ (often ≥ 1), raising the physical thermal noise floor above the ideal prediction.

3.2 2. Total Noise and Flicker Corner

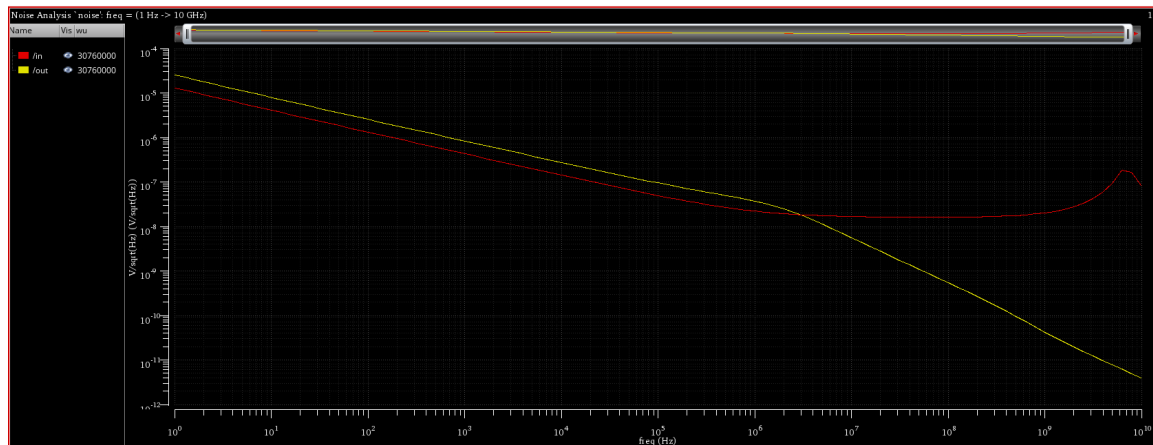


Figure 15: Total Input/Output Noise (log scale) to get flicker corner

Flicker Noise Corner Estimation

Q: Estimate the Flicker noise corner from the total noise plot.

The flicker ($1/f$) noise corner frequency occurs at approximately $f_c = 773$ kHz, which is the intersection point where the flicker noise asymptote meets the flat thermal noise floor.

4. Part 4: 5T OTA Transient Noise Analysis

4.1 1. Buffer Verification and Transient Noise

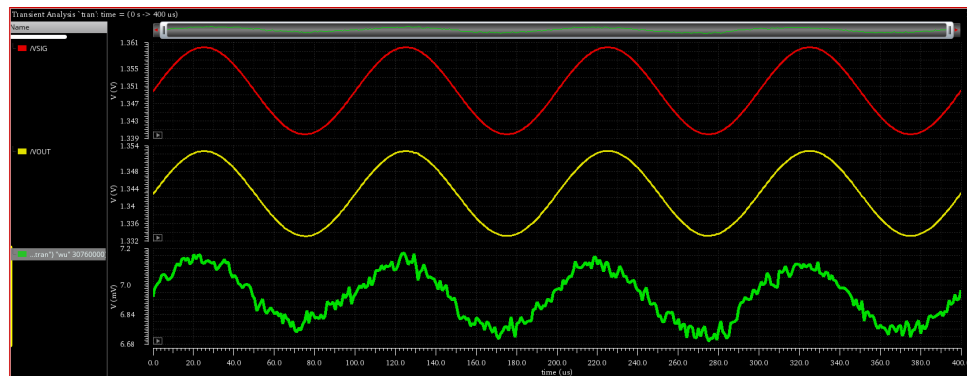


Figure 16: Input Vsig, Vout and superimposed noise vs time at $F_{IN}=10\text{kHz}$

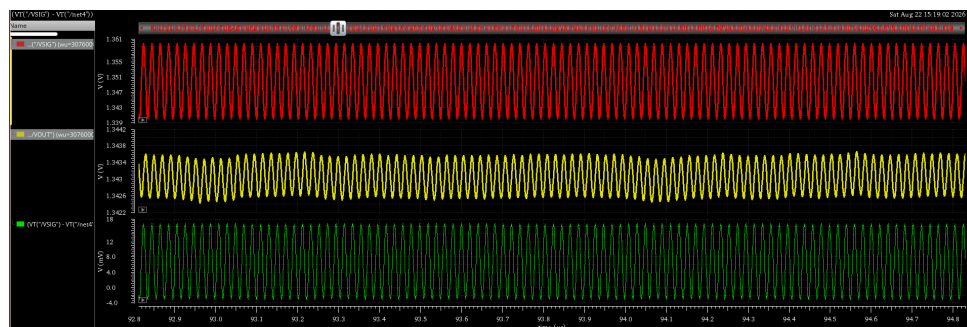


Figure 17: Input Vsig, Vout and superimposed noise vs time at $F_{IN}=48.9\text{MHz}$

4.2 2. Multiple Runs Analysis (20 Iterations)

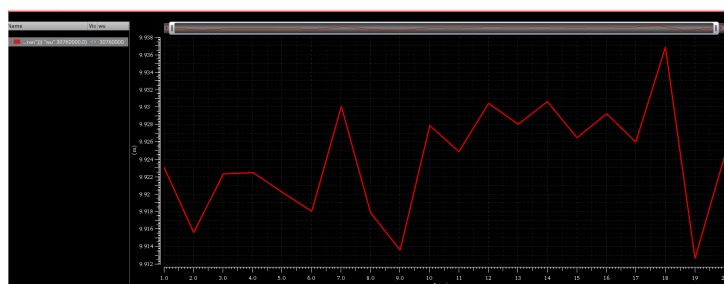


Figure 18: RMS Noise per iteration

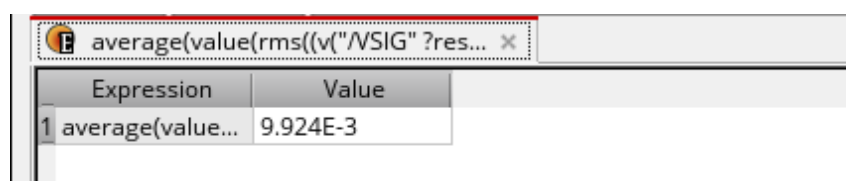


Figure 19: Average RMS